

Midterm Review

What is HCI?

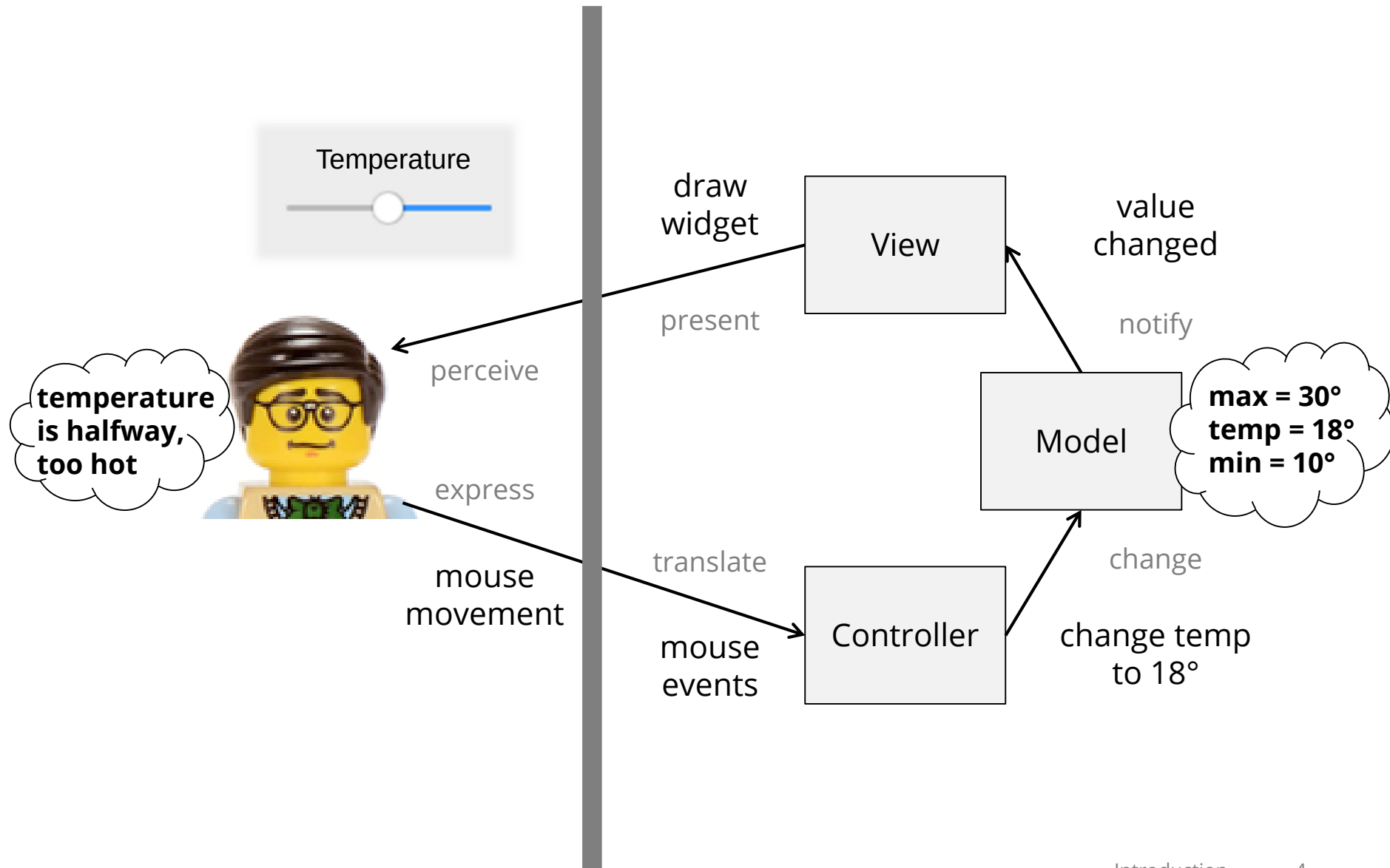
- “Human-computer interaction is a discipline concerned with the **design, evaluation and implementation of interactive computing systems** for **human use** and with the study of major phenomena surrounding them”.

-ACM definition

Interface vs. Interaction

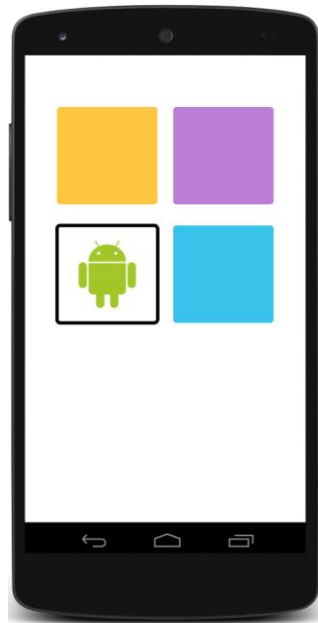
- What is the difference between an *interface* and *interaction*?
- **Interface** refers to the external presentation to the user
 - Controls (what you can manipulate to communicate intent)
 - Feedback (what the program uses to communicate its response)
- **Interaction** refers to actions by user and system over time
 - interaction is a dialog with a cycle alternating between the user manipulating controls and the system responding with feedback

Graphical Temperature Control

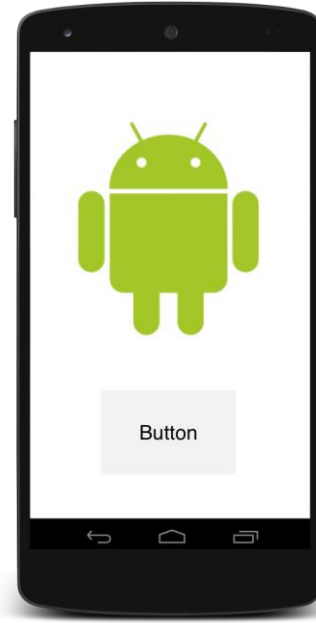


Activities

- A standard application component is an Activity
 - Typically represents a single screen
 - Main entry point (equivalent to main() method)
 - For most purposes, this is your application class



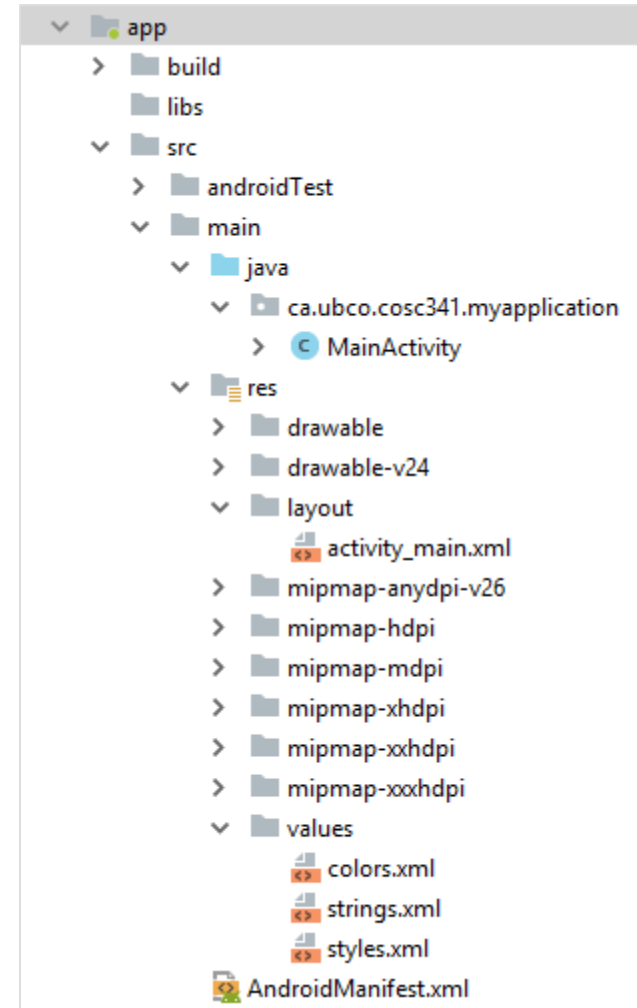
Activity 1

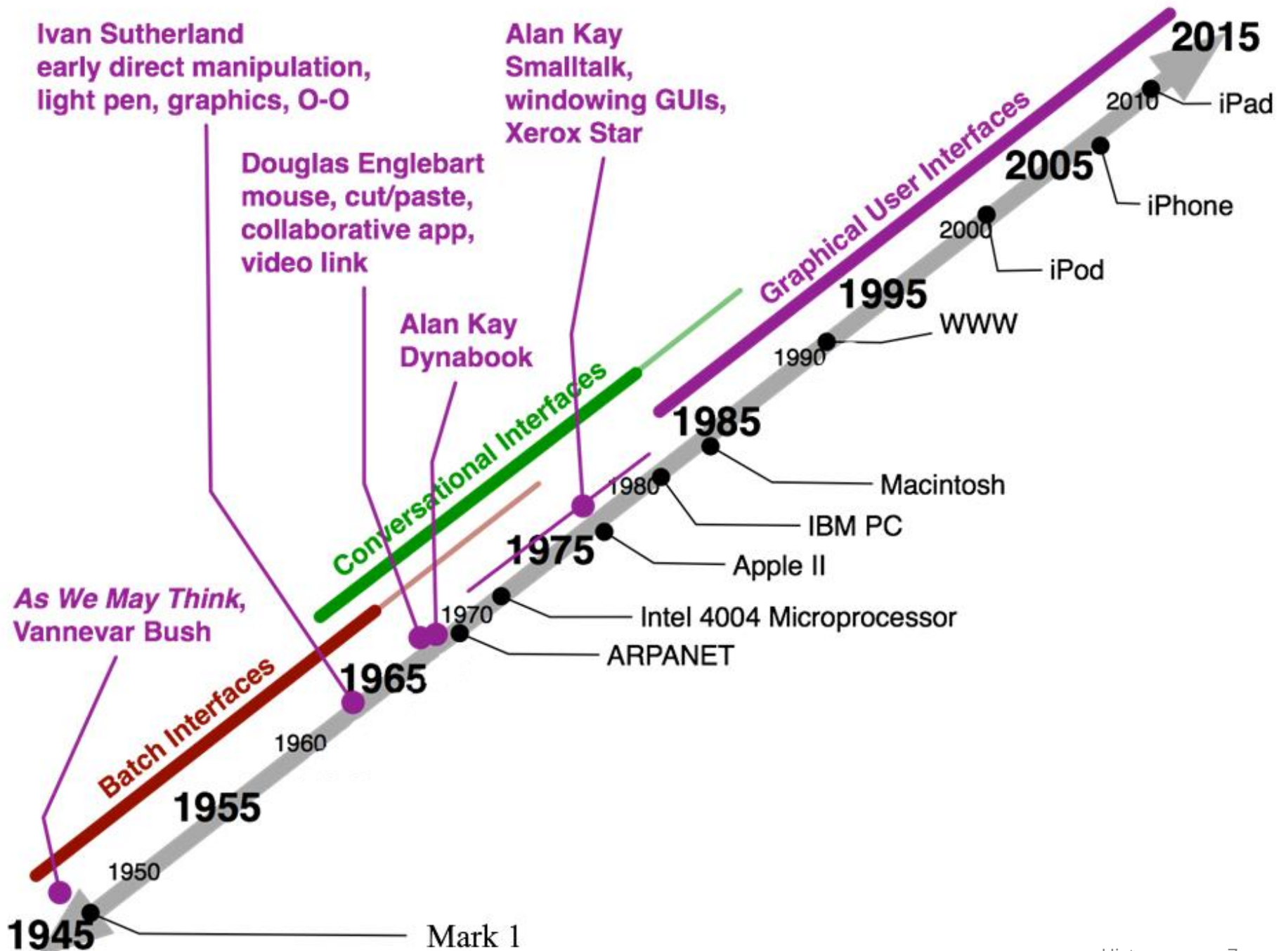


Activity 2

Android Project Files

- Manifest (app/manifests/)
 - Application setting
- Java (app/java/)
 - **(*.java) source code**
- Resources (app/res/)
 - **layout: (*.xml) UI layout and View definitions**
 - **values: (*.xml) constants** like strings, colours, ...
 - also bitmaps and SVG images (mipmap*, drawable*,)





Usability

- Usability is a quality attribute that assesses how easy user interfaces are to use
- Some usability goals include:
 - Effectiveness
 - Efficiency
 - Safety
 - Utility
 - Learnability
 - Memorability

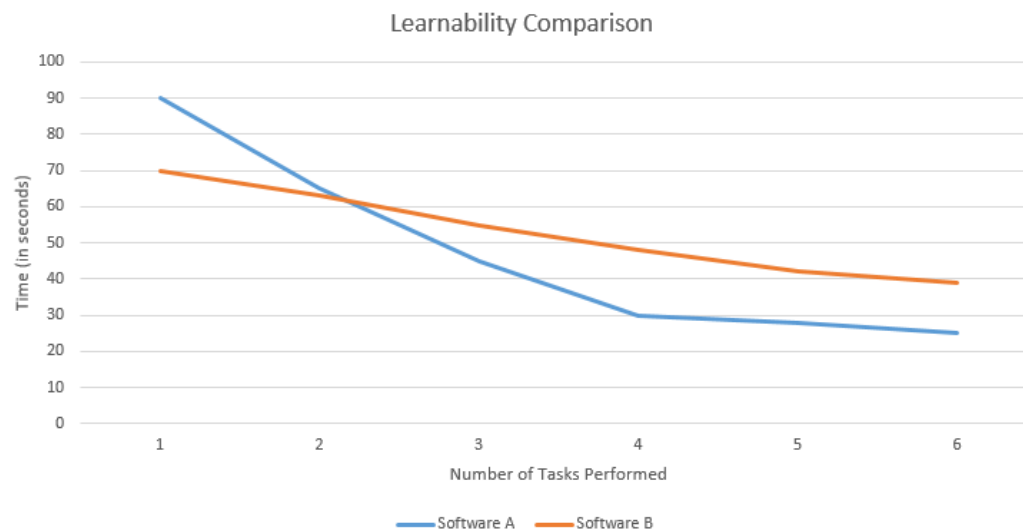
<https://www.nngroup.com/articles/usability-101-introduction-to-usability/>

Usability Goals

- Efficiency
 - Effective, ok; but efficient?
 - Considers how much time/number of steps it will take users to perform their tasks
 - Systems that require the least time/number of steps are usually considered as being efficient
 - Criteria: time to complete a task; # of operations to complete a task

Example Usability Goal

- Learnability
 - important if system will be adopted by the user
 - A more learnable system is one that reduces the time it takes to complete tasks as users spend more time with a system faster than others
 - Question: can primary (core) and secondary tasks be learned quickly and easily?
 - Criteria: time to learn a task, errors made in learning a task
e.g. smartphone?



User Experience Goals

- A set of user experience goals include:
 - Satisfying: productive
 - Motivating: did not feel like giving up
 - Enjoyable: no frustrations were encountered
 - Aesthetically pleasing: subjective
 - Fun: excited about using it again
 - Supportive of creativity: drawing tools
 - Entertaining: games
 - Rewarding: sense of productivity
 - Helpful: clueless but still made it through

Generic Design Principles

- Visibility
- Feedback
- Constraints
- Mapping
- Consistency
- Affordance
- Simplicity
- Matching
- Minimizing memory load
- Diagnose/Recover errors
- Control and freedom
- Flexibility
- Provide Help

Categories of users

- Three categories of users:
 - Primary:
Frequent hands-on (everyday)
 - Secondary:
Occasional or via someone else (getting reports)
 - Tertiary:
Affected by its introduction (owner)

User Profile

- Collection of attributes for describing a collection of users
 - i.e., age, disabilities, gender, culture, experience, plus any other factors relevant to a particular system
 - E.g., ages 18-35
- A user profile will help you to understand who you are building your product for
- Most systems will have a number of different profiles

Persona

- Personas
 - Personas are **not real people**, but they represent them through the design process
 - A number of **fictitious users** with specific values for all of the relevant attributes
 - Rich descriptions of typical users
 - Make profiles more life-like

Requirements

- Five primary types:
 - Functional
 - Data
 - Environmental / Context of Use
 - User
 - Usability

Data Gathering Involving Users

- Primary methods
 - Interviews
 - Questionnaires
 - Observations
 - Focus groups
- Advantages and disadvantages of each method

Interviews



Questionnaires



Observation



Focus Group



Overview of techniques

Technique	Purpose	Data type
Interviews	Exploring issues	Mostly Qualitative
Questionnaires	Answering specific questions	Qualitative and Quantitative
Observation	Understanding context of user activity	Qualitative
Focus Group	Collecting Multiple Viewpoints	Mostly Qualitative

What is a Prototype?

- In interaction design it can be (among other things):
 - a series of screen sketches, i.e. not necessarily related
 - a storyboard, i.e. a cartoon-like series of scenes
 - a Powerpoint slide show, e.g., simulation
 - a video simulating the use of a system
 - a cardboard mock-up
 - a piece of software with limited functionality written in a language
- Key point:
 - Goal is to NOT the artifact, but the feedback

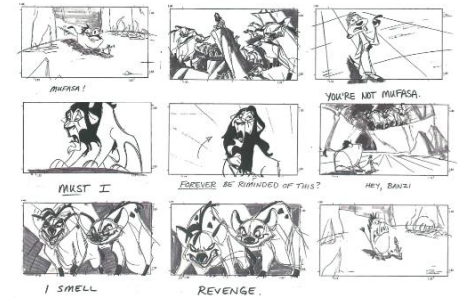
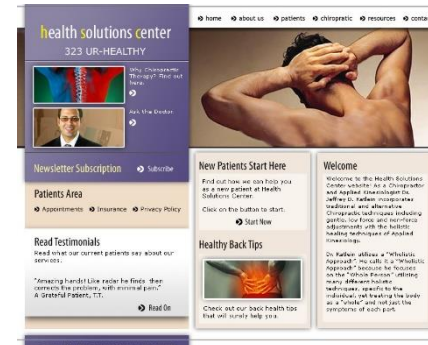
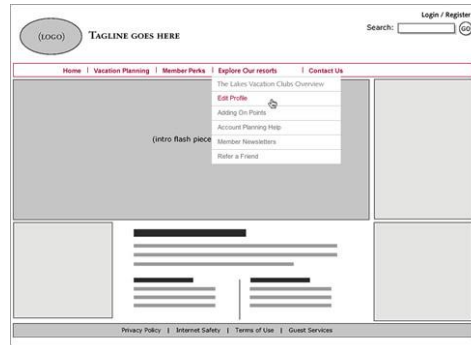
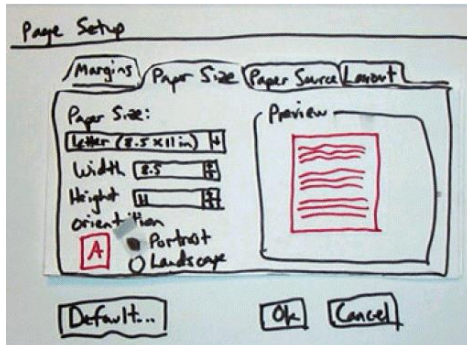


Image source: Nintendo

Prototype Fidelity

There are many types of prototypes, ranging anywhere between these two extremes:

- Low fidelity: prototype doesn't look much like final product, operation of the prototype may be simulated and slower.
- High fidelity: prototype looks and operates like the real product.



Why Prototype?

- **Communication:** discuss ideas with stakeholders
 - “Where’s the ON button?”
- **Evaluate** interface effectiveness for communicating conceptual model
 - “Whoops, users didn’t understand that.”
- Develop **requirements** and/or specifications •
 - “Uh-oh, here’s something we forgot.”
- **Learning** and problem solving
 - “Hey, that will work!”
- Further develop **conceptual and physical design**
 - “That’s way too heavy”
- **Save time and money**
 - Don’t waste time coding/building the wrong thing

What to Prototype?

- **Screen layouts and information display:** placement of different widgets, types of widgets, etc
 - e.g. should edit fields come after/before control buttons
- **Work flow, task design:** the sequence of steps required to achieve the task
 - e.g. flow from one screen to another, changes on the screen
- **Difficult, controversial, critical areas:** provide alternative solutions to be assessed
 - e.g. business critical reports with specific alignment of fields
- **Technical issues:** will the graphics rendered be fast enough?

When to prototype?

- To get out of an **early**, focused discussion, and to reach agreement
- when you need to **communicate ideas**
 - design team, managers, users etc.
- when you have questions and you can't proceed:
 - **functionality:**
structure, sequencing, flow, clarity & completeness of information
Example: knob versus slider for controlling volume
 - **appearance:**
branding, clarity, aesthetics, color, shape, etc.
screen too crowded? actions clear, or lost in clutter?
 - **specifications:**
“design by prototyping” (evolutionary approach)
how many files to show in file selection box

Prototypes

- Storyboarding/Paper prototype benefits
- Storyboarding/Paper prototype limitations
- What can/can't be learned from low/high-fidelity prototype
- Pros and cons of low/high-fidelity prototype
- Video prototype, wizard-of-oz

Limiting Functionality: How Much to Represent?

- Vertical prototypes
 - includes **in-depth functionality** for only a few selected features
 - common design ideas can be tested in depth
- Horizontal prototypes
 - surface layers includes the **entire user interface** with **limited underlying functionality**

